Formula Slug - Power Electronics Onboarding

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1 Introduction

Hi, my name is Joseph, and I'm a 3rd year EE major. This project has been super fun, I've already learned quite a bit from it! FS sounds really exciting and I really hope to gain a chance to learn more and build practical skills in engineering.

2 Calculations

The buck converter is designed for 12 V input, 5 V & 2 A output. Using the equation:

$$V_{out} = DV_{in}$$

We find that we want a duty cycle of $D = 0.4167$, where the switch is on approximately 41.67% of the time.

I chose a switching frequency of $f_s = 700$kHz, which is above the 500kHz specified in the document. I went higher mainly because I wanted to decrease the size of the inductor.

I aimed for an output voltage ripple of 1% and inductor current ripple of 30%, giving:

$$\Delta V_{out} = 0.05; \quad \Delta I_{out} = 0.6$$

To calculate the component values, I used the buck converter formulas:

$$L = \frac{(V_{in} - V_{out}) \cdot D}{\Delta I_L \cdot f_s} = \frac{(12-5) \cdot 0.4167}{(0.6)(7 \times 10^5)} \approx 6.95\,\mu\text{H}$$

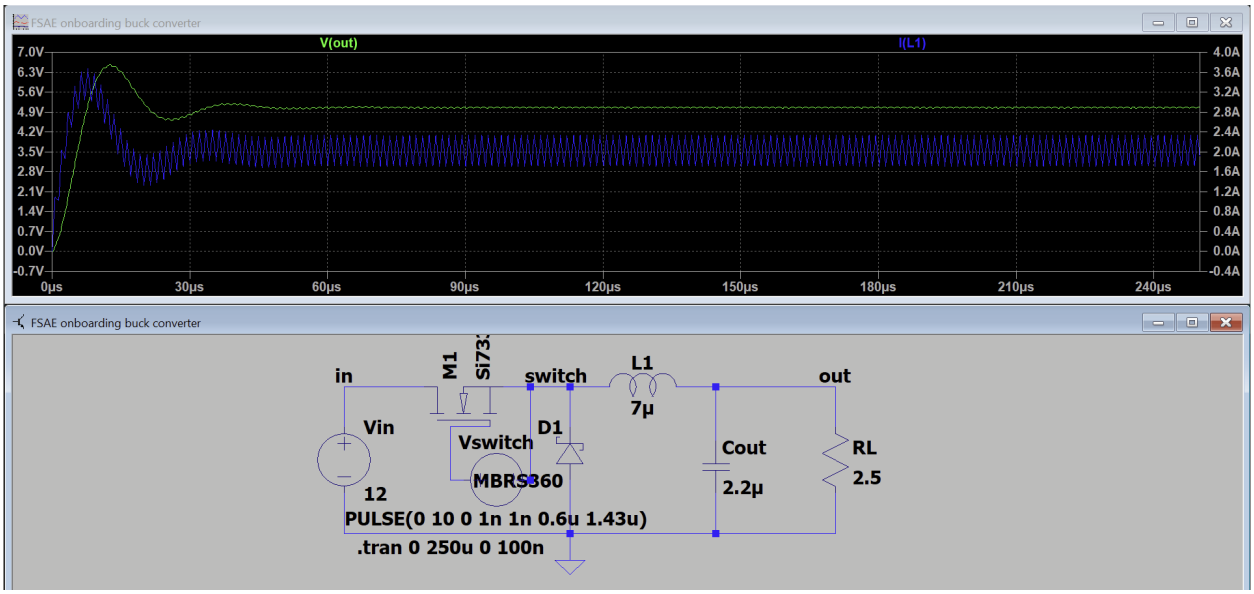
$$C = \frac{\Delta I_L}{8 f_s \cdot \Delta V} = \frac{0.6}{8(7 \times 10^5)(0.05)} \approx 2.14\,\mu\text{F}$$



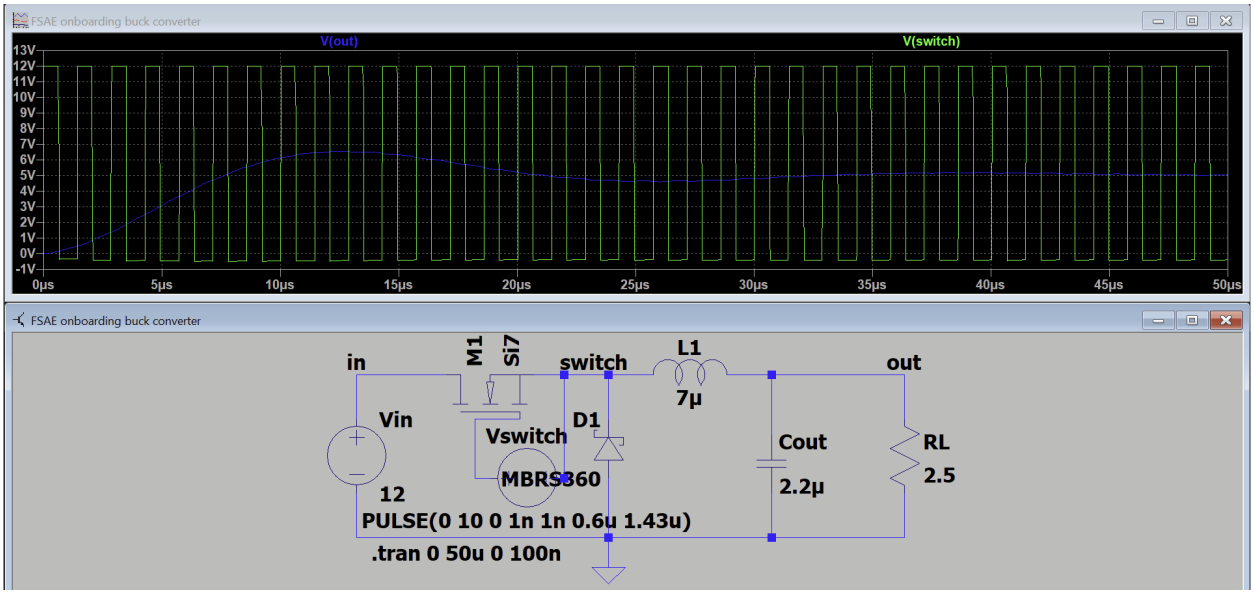
3 Simulation

I have never used SPICE software before, so I figured this was a good time to learn the basics... I wanted to make sure the thing works at least in theory

I constructed the converter topology in LTspice and added a load of 2.5 Ω for a 2 A output. In order to simulate the switching, I used an NMOS connected to another voltage source set to PWM for the duty cycle.



It was also pretty cool to observe the transient starting behavior:



4 Sourcing

Component	Cost	Link	Units	Size
Buck IC	\$2.14	TPS5405DR	1	5.0 x 6.19 mm
7 μH Inductor	\$0.75	DR1040-7R0-R	1	10.5 x 10.3mm
2.2 μF Capacitor	\$0.16	GRM188R71A225KE15D	1	1.6 x 0.8 mm
10 nF Capacitor	\$0.16	0603B103K101SD	1	1.6 x 0.8 mm
33 kΩ Resistor	\$0.10	ERJ-2GEJ333X	1	1.0 x 0.5 mm
Schottky Diode	\$0.89	MBRS360T3G	1	5.6 x 3.95 mm
Total Cost	\$4.20			

For the buck switching IC, I selected TPS5405DR by Texas Instruments. It's rated for a maximum input of 28V, min/fixed output of 5V, and current output of 2A. It has a frequency switching range from 50kHz to 1.1 MHz, configurable by connecting one of its pins with a resistor to ground. A graph provided in the data sheet shows that a resistance of about 33 kΩ will result in a 700 kHz switching frequency.

It also requires external slow start, and with the formula and values provided: $V_{ref} = 0.8\,\text{V}$, $I_{ss} = 0.2\,\mu\text{A}$, a recommended $t_{ss} = 1 - 10$ ms, and

$$C_{ss} = \frac{I_{ss} t_{ss}}{V_{ref}} = \frac{(2 \times 10^{-}5)(0.005)}{0.8} \approx 13\,\text{nF}$$

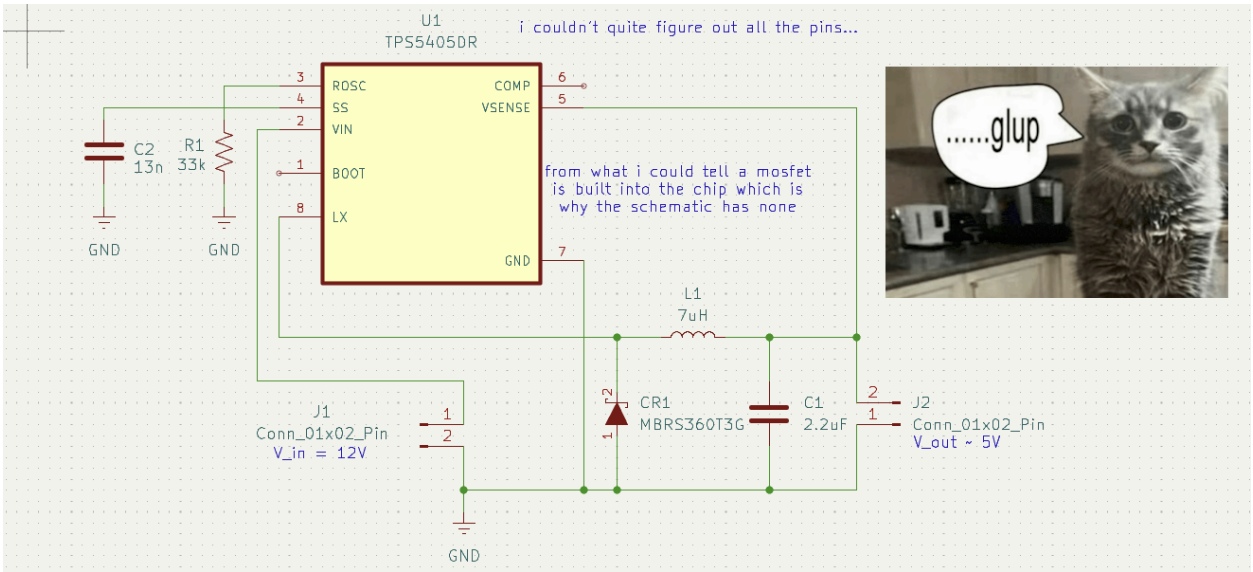
I chose an a 7 μH inductor, very close in value to the calculation of 6.94 μH. It's rated for a current of 4.5 A and saturates at 4.8 A, well above the ripple observed in the simulation (approx. 1.7 A - 2.3 A). The downside is the inductor I chose appears to be fat as shit, which I realized when attempting to lay out the PCB.

I chose an output capacitor of 2.2 μF, which is close in value to the 2.14 μF calculated previously. It's rated for 10 V, which is greater than the 5 V load that the capacitor is in parallel with.

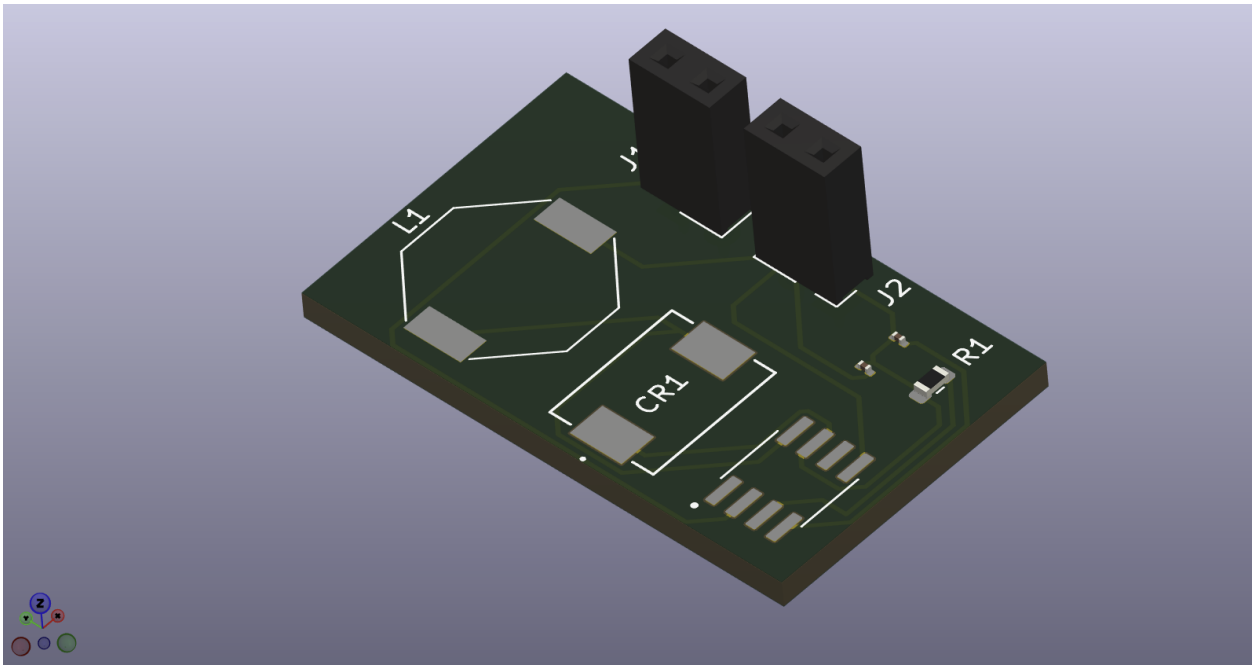
The diode I was very unsure of choosing because this thing is pretty fucking big. My simulation showed a current of nearly 3 A, which is what I chose the diode based off of. However, I realize that I'm not exactly sure what the 3 A average rectified current specification means.

5 Schematic & PCB

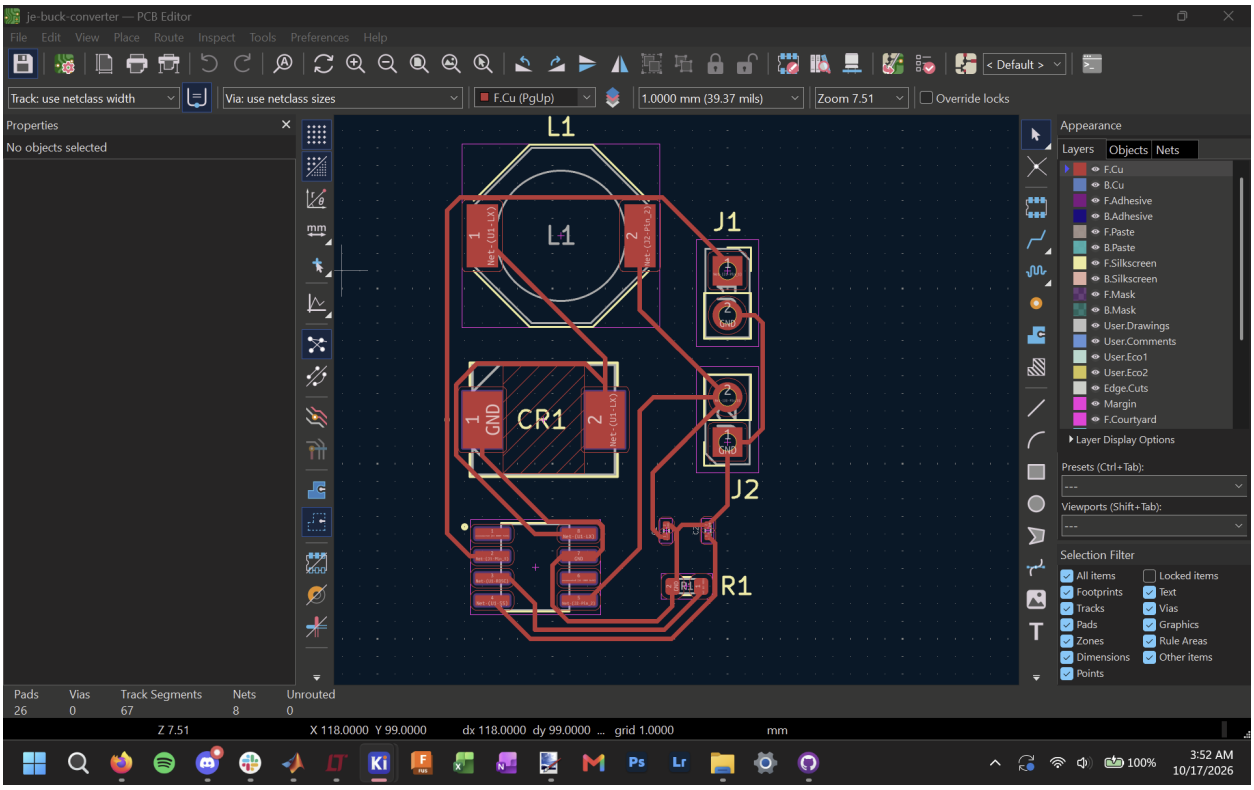
i didn't finish... glup...
I didn't quite understand what a couple of the pins did, and I guess I also kind of skipped the input filter capacitor



I also could not figure out why some of my components had disappeared on the PCB rendering. I checked the design constraints like it said, but I still didn't know why.



I have never used EDA or designed a PCB before, so I was literally just throwing shit down and trying to connect everything. I couldn't find every component exactly and had to import the IC and the diode, and the inductor is just one that's approximately the same size. I also didn't know what connectors would be used, so I used one that looks familiar and may not be right.



6 Reflection

This was a pretty fun project and I enjoyed learning some basics of power electronics. I also learned some stuff about component selection and simulation, which I know will be helpful to me in the future. I felt I was glossing over other constraints like noise and heating, which I also hope to gain experience in dealing with. Overall, I felt this was a pretty fulfilling experience.